

Notes for R H Bing's Plane Topology Course

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Introduction by Sam Creswell

R H² Bing (1914 – 1986), a native of Oakwood³, Texas, was a 20th century mathematician and teacher of mathematics.

Bing received a bachelor's degree from Southwest Texas State Teachers College (now, Texas State University - San Marcos) and began teaching and coaching football in Texas public schools. He took a summer course under R. L. Moore, who recognized his immense talents, encouraged him to work toward a doctorate, and provided a teaching position for him while he was in graduate school. Upon receipt of his doctorate from Moore, Bing accepted a position at Wisconsin and soon cemented his reputation as a teacher and researcher. Late in his career, Bing returned to Texas, where he served as the chair of the department, a position which he once held at Wisconsin.

Bing served as a President of both the Mathematical Association of America and the American Mathematical Society. Further, he was member of the National Academy of Sciences.

In common with his mentor, R. L. Moore, Bing was passionately interested in undergraduate mathematics. As Moore found immense talent in a Texas high school football coach, Bing was also interested in recognizing and developing talent. Few mathematicians with Bing's prominence attempt to discover talent among those whose mathematical background is modest. These notes manifest his interest in doing so.

A partial list of his accomplishments may be found on page 220 of [1]. Further, [1] provides much more information about Bing, his mentor, and inquiry-based learning in mathematics, as implemented at Texas.

As these notes passed through the hands of many instructors and typists over the years, a handful of typographical errors naturally developed. In consultation with the editors of this journal, we have made corrections, as necessary.

Sam Creswell
Mathematics Department

²Bing had no given names, as such. His first name was the letter 'R' and his second name was the letter 'H' (see p. 208 [1]). Bing once directed some painters at Wisconsin to produce some signage for him writing down 'R only H only Bing'. They duly produced signage saying 'Ronly Honly Bing'!

³Visits to Oakwood support the observation that mathematical talent may be found anywhere. The Bings who live in Oakwood today are aware of their kinsman's prowess and are proud of his achievements.

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Introduction to the Student

In this course we study point sets in the plane, properties of these sets, and operations with these sets. Special emphasis will be placed on the aspects of plane topology that are used in calculus.

Rather than trying to consider many theorems superficially, only a selected group will be studied. It is felt that if a person understands these, he can prove many others.

Before proofs are given in class, the student will be given all opportunity to work them out himself. The more proofs that the student can develop himself, the more mathematical training he obtains. In fact, much original research has developed because a mathematician, in developing for himself the proof of a known theorem, finds new methods that can be used elsewhere. In proving a theorem a person may learn much more than just the steps in the proof. He may learn which methods and tricks will work and which will not (this is important too) in similar situations.

Mathematics may be thought of as a way of thinking rather than a collection of facts. The person who can develop a proof for a theorem or work out a solution of a problem is regarded more highly as a mathematician than the person who only remembers many mathematical facts and names of various theorems.

In this course, emphasis will be placed on mathematical thinking. Although there will be many mathematical truths developed, importance will be given to the development rather than to the facts themselves. This is not to say that facts of mathematics are not important. They are tools that enable us to do a better job of mathematical thinking and should be learned well.

Chapter 1

Operations with sets

We need to acquaint ourselves with the following terminology, undefined terms, and definitions of topology.

Point. In synthetic geometry and in topology, a point is usually regarded as an undefined term.

In this course, we suppose that the points under consideration lie in the Euclidean plane (unless specified otherwise). We will feel free to use the properties of the plane learned in plane geometry and analytics.

In courses where the number system is developed first, a point is sometimes defined in terms of numbers. For example, in analysis, a point may be defined to be a complex number; in plane analytics, a point is an ordered pair of real numbers; in 3-dimensional Euclidean analytic geometry, a point is an ordered triple of numbers; and in 4-dimensional Euclidean analytic geometry, a point is an ordered quadruple of numbers.

Definition 1.1. A **point set** is a collection such that each element of the collection is a point.

If a collection contains only one element, there is a distinction between the collection and the element, though for brevity, we do not always emphasize this.

Definition 1.2. The **null set** is a collection with no element.

There is some objection to introducing such an artificial set. We have no need for null sets in this course and in places where non-null is mentioned we could state the results more simply if the convention were followed that each set contains a point. However, this convention is not generally followed.

Definition 1.3. $p \in X$ means that p is an element of the collection X . If p is not an element of X , we write $p \notin X$.

Definition 1.4. $X \subset Y$ means that each element of X is an element of Y . If

$X \subset Y$, we say that X is a **subset** of Y . If X is a subset of Y but X is not all of Y , we say that X is a **proper subset** of Y . See Figure 1.1.

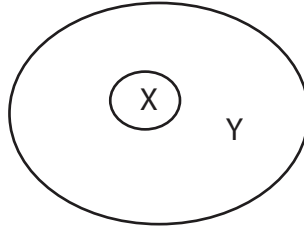


Figure 1.1: X is a Subset of Y .

Definition 1.5. $X \cdot Y$ or $X \cap Y$ (the **intersection**, **product**, or **common part** of X and Y) denotes the set of all points in both X and Y ; that is, $X \cdot Y = \{p \mid p \in X \text{ and } p \in Y\}$. See Figure 1.2.

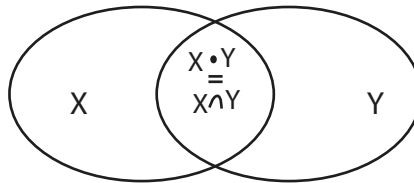


Figure 1.2: The intersection of X and Y .

Definition 1.6. $X + Y$ or $X \cup Y$ (the **sum** or **union** of X and Y) denotes the set of all points in either X or Y . That is, $X + Y = \{p \mid p \in X \text{ or } p \in Y\}$. See Figure 1.3.

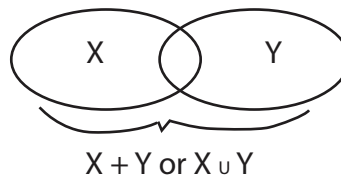


Figure 1.3: The sum of X and Y .

Definition 1.7. $X - Y$ (the *difference* of X and Y) denotes the collection of points of X that do not belong to Y . That is, $X - Y = \{p \mid p \in X \text{ and } p \notin Y\}$. See Figure 1.4.

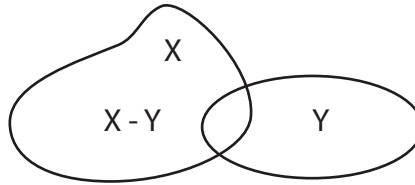


Figure 1.4: X Minus Y .

Chapter 2

Limit Points

Definition 2.1. A *neighborhood* is the interior of a circle.

Definition 2.2. A *neighborhood of a point* is a neighborhood containing the point. We do not suppose that the point is the center of the neighborhood.

As an extension of the notion of a neighborhood in other spaces with a distance function, a neighborhood is regarded as the interior of a generalized sphere.

Definition 2.3. The ε -*neighborhood of p* (written $N(p, \varepsilon)$) is the set of all points whose distance from p is less than ε .

We denote the distance between p and q by $\rho(p, q)$. Then $N(p, \varepsilon) = \{q \mid \rho(p, q) < \varepsilon\}$. If a point r lies in $N(p, \varepsilon)$, then $N(p, \varepsilon)$ is a neighborhood of r , although it may not be the ε -neighborhood of r . See Figure 2.1.

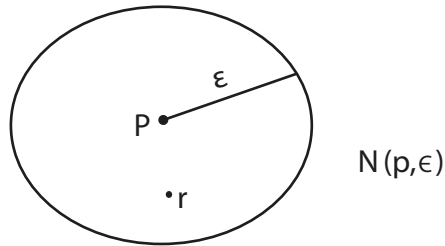


Figure 2.1: ε -Neighborhood of p .

If a space does not have a distance function, certain point sets in it may be designated as neighborhoods. A neighborhood of p is one of those designated point sets that contains p . These neighborhoods must satisfy certain conditions called axioms. Two axioms sometimes used are the following:

Axiom 1. If U and V are neighborhoods of the same point p , there is a neighborhood W of p such that $W \subset U \cdot V$.

Axiom 2 (Hausdorff or T_2 separation axiom). If p and q are different points, there are neighborhoods N_p and N_q of p and q respectively such that N_p does not intersect N_q .

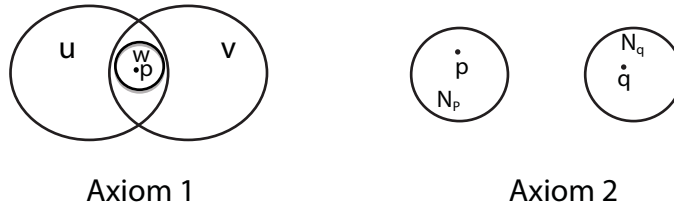


Figure 2.2: Axioms 1 and 2.

See Figure 2.2 for illustrations of these axioms. Sometimes weaker axioms are used but more frequently, stronger ones are employed.

Definition 2.4. If N is a neighborhood of a point p , we call $N - \{p\}$ a **deleted neighborhood of p** . Here $\{p\}$ denotes the collection whose only element is p .

Definition 2.5 (Open set). A point set D is **open** if each point p of D has a neighborhood that lies in D .

An open set can be expressed as a sum of neighborhoods, although the neighborhoods may be infinite in number. The interior of a square is open, but a disk (a circle plus its interior) is not.

Definition 2.6 (Neighborhood definition of limit point). The point p is a **limit point** of the point set M if each deleted neighborhood of p contains a point of M .

Definition 2.7 (ϵ -definition of a limit point). The point p is a **limit point** of the point set M if for each positive number ϵ there is a point q of M such that $0 < \rho(p, q) < \epsilon$.

Definition 2.8. The **boundary** of a set X is the set of all points p such that each neighborhood of p contains both a point of X and a point not of X .

Definition 2.9. The **closure** of a set X is the sum of X and all its limit points. This closure is designated by $\overline{X}$.

It follows from these definitions that the closure of X is equal to X plus its boundary.

Definition 2.10. A **closed set** is one that contains all its limit points.

Definition 2.11. The **complement** of a point set X is the collection of all points not in X .

2.1 Theorems

Prove the following theorems for the plane.

1. The sum of two open sets is open.
2. If p is a limit point of the sum of two closed sets, it is a limit point of one of them.
3. The sum of two closed point sets is closed.
4. The product of two closed sets is closed.
5. If a point lies in each of two neighborhoods, it lies in a neighborhood in the intersection of the two neighborhoods.
6. The product of two open sets is open.
7. If p is a limit point of A , each neighborhood of p contains infinitely many points of A .
8. For each set X , $\overline{X}$ is closed.
9. For each set X , $\overline{X} = \overline{\overline{X}}$.
10. The boundary of a set is closed.
11. If p is a limit point of $A \cdot B$, it is both a limit point of A and a limit point of B .
12. The sum of a collection of open sets is open.
13. The intersection of a collection of closed sets is closed.
14. The closure of a set X is the intersection of all closed sets containing X .
15. The complement of a closed set is open.
16. The complement of an open set is closed.
17. The complement of $X + Y$ is the intersection of the complement of X and the complement of Y .
18. The complement of $X \cdot Y$ is the sum of the complement of X and the complement of Y .

2.2 Questions

Answer the following.

1. If X is a finite set of points on the x -axis, does X have a limit point?
2. If X is a subset of the x -axis and p is a limit point of X , does p belong to x -axis?
3. Is the interior of a circle the sum of an infinite collection of closed sets?
4. Name a property that is not possessed by the sum of some two sets even if it is possessed by each of them.
5. Note that Theorem 7 does not state that there are infinitely many points that belong to each neighborhood. Explain why there may not be even one such point.
6. Why may the sum of a collection of closed sets fail to be closed?
7. Why may the intersection of a collection of open sets fail to be open?
8. Name a plane set that is both open and closed. Name one that is neither.
9. Consider the set $M = \{(x, y) \mid 0 < x^2 + y^2 < 1 \text{ or } x = y = 1\}$. Which of the points $(0, 0)$, $(0, 1)$, $(1, 1)$, $(2, 0)$, and $(1/2, 0)$ are points of M , and which of them are limit points of M ?
10. If each of three sets contains a point of $A + B$, does this mean that either each of them contains a point of A or each of them contains a point of B ?
11. Which of the theorems in the first set shows that if p is not a limit point of X and is not a limit point of Y , then it is not a limit point of $X + Y$?
12. Show that plane neighborhoods satisfy the two axioms mentioned for abstract neighborhoods.

Chapter 3

Connectedness

Intuitively, we think that a set is connected if it is “all in one piece.” We use the following definitions.

Definition 3.1. Two sets are *disjoint* or *mutually exclusive* if neither contains a point of the other.

Definition 3.2. Two sets are *mutually separated* if neither contains either a point or a limit point of the other.

Each of the two mutually separated sets is relatively open and relatively closed in their sum.

Definition 3.3. A set A is *relatively open (relatively closed)* in Y if A is the intersection of Y and an open (closed) set.

A circle and its interior are mutually exclusive but they are not mutually separated. However, the interior of a circle and the exterior are mutually separated.

Definition 3.4. A set is *not connected* if it can be expressed as the sum of two non-null mutually separated sets.

Definition 3.5. A set is *connected* if it is not the sum of any two non-null mutually separated sets. See Figure 3.1.

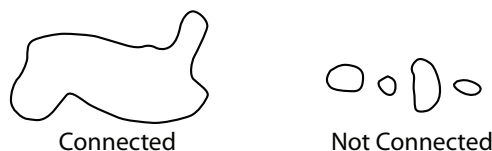


Figure 3.1: Connected vs. Not Connected

Definition 3.6. A *component* of a set X is a maximal connected subset of X . That is, it is a connected subset of X that is not properly contained in any larger connected subset of X .

3.1 Theorems

Prove the following.

19. If X is connected and Y is a subset of the boundary of X , then $X + Y$ is connected.
20. If a point p belongs to each element of a collection of connected point sets, their sum is connected.
21. If p is a point of the connected set M and $M - p$ is the sum of the mutually separated sets H and K , then $H + p$ is connected.
22. If p is a point of a point set X , then some component of X contains p .

3.2 Exercises

1. Show that the intersection of two connected sets may fail to be connected.
2. Is any finite set connected?
3. Name a property such that there is no set which is maximal with respect to having the property.

Chapter 4

The Real Line

Let $\mathbb{R}$ be the number system with its ordinary topology. Then topologically $\mathbb{R}$ is like a line. Here are three statements about $\mathbb{R}$ that are of utmost importance.

1. $\mathbb{R}$ is connected.
2. Each bounded monotone increasing sequence of elements of $\mathbb{R}$ has a least upper bound.
3. $\mathbb{R}$ satisfies the Dedekind cut property.

None of these statements about $\mathbb{R}$ would be satisfied if we used the convention that only rational numbers are real. Hence we need an axiom of some sort in order to prove the properties. The one that we adopt is the following.

Axiom 3. *A straight line is connected.*

Definition 4.1. A set X in $\mathbb{R}$ is **bounded** if there is a positive number m such that each element of X lies in the m -neighborhood of 0.

Definition 4.2. A sequence of numbers $A_1, A_2, \dots$ is **monotone increasing** if each term of the sequence is larger than the one before it. **Monotone decreasing**, **monotone nonincreasing**, and **monotone nondecreasing** are defined in a similar fashion.

Definition 4.3. An **upper bound** of a collection X is a number that is greater than or equal to each element of X . The least such upper bound is called the **least upper bound** of X . Similarly, a **greatest lower bound** may be defined.

Definition 4.4. If X has a largest value, this value is called the **maximum** of X . If there is a value of X that is the largest value in some neighborhood, it is called a **relative maximum**. Similarly, a minimum may be defined.

Definition 4.5. The *Dedekind cut property* says that if $\mathbb{R}$ is expressed as the sum of two non-null, mutually exclusive sets A and B , such that each element of A is less than each element of B , then either A has a maximum or B has a minimum.

Definition 4.6. If p and q are two points of a straight line, the *interval* $[pq]$ is the sum of p , q , and all points between them. We use (pq) to denote the *open interval* or *segment*. $(pq) = [pq] - (\{p\} + \{q\})$. Both $[pq]$ and (pq) are called *sects*.

4.1 Theorems

Prove the following.

23. A segment is connected. (Use Axiom 3 to show this.)
24. An interval is connected.
25. Each bounded monotone increasing sequence of numbers has a least upper bound. (Use Axiom 3 to show this. Theorem 25 itself is taken as an axiom in some calculus books.)
26. $\mathbb{R}$ satisfies the Dedekind cut property.

4.2 Questions

1. Is the set of all points on the x -axis with rational abscissas (x values) connected?
2. If X is a bounded sequence of numbers and Y is the set of all numbers that follow infinitely many elements of X , is Y open, closed, or neither?

Chapter 5

Sequential Compactness

Definition 5.1. A set M is **sequentially compact** if for each infinite subcollection of M there is a point p of M which is a limit point of this subcollection.

Definition 5.2. A set is **bounded** if it lies on the interior of some circle.

Definition 5.3. A sequence of sets $A_1, A_2, \dots$ is called a **nested sequence** if for each positive integer i , $A_{i+1} \subset A_i$. A point is called a **common point** if it belongs to each A_i .

Suppose $f(x)$ is defined on an interval $(a \leq x \leq b)$. The notion of sequential compactness may be used in proving the following useful theorems from calculus which hold if f is continuous. See Chapters 9 and 10.

1. f is bounded.
2. f takes on a maximum and a minimum.
3. f is uniformly continuous.
4. $\int_a^b f(x)dx$ exists.

5.1 Theorems

Prove the following.

27. A sequentially compact set is closed.
28. An interval is sequentially compact. (Use Axiom 3 to show this.)
29. A square plus its interior is sequentially compact.
30. A bounded closed set is sequentially compact.
31. A sequentially compact set is bounded.

32. A closed bounded set has a highest point.
33. If p is a point and M is a closed set (non-null), there is a closest point of M to p .
34. (Bull's eye theorem). Each nested sequence of non-null sequentially compact sets has a common point.

5.2 Questions

1. Is the plane sequentially compact? The surface of a sphere?
2. Does each nested sequence of segments have a common point?

Chapter 6

Countable Collections

Definition 6.1. A collection is called *countable* if it has only a finite number of elements or if there is a one-to-one correspondence between its elements and the positive integers. A collection that is not countable is called *uncountable*.

6.1 Exercises

1. Give a plan for putting the rational numbers in a one-to-one correspondence with the positive integers.
2. Set up a one-to-one correspondence between the positive integers and the odd positive integers.
3. Set up a one-to-one correspondence between the points of a long interval and the points of a short interval.
4. Show that if M is a closed proper subset of an interval I that contains both ends of I , then $I - M$ is the sum of a countable collection of segments.
5. Does each uncountable subset of the x -axis contain a segment?

6.2 Theorems

Prove the following.

35. The rational numbers are countable.
36. The numbers between 0 and 1 are uncountable.
37. Each infinite collection has uncountably many subsets.
38. The sum of a countable collection of countable sets is countable.

39. No countable set with more than one point is connected.

Chapter 7

Coverings

Definition 7.1. A collection G of point sets is said to **cover** a set A if each point of A is contained in an element of G .

Definition 7.2. An **open cover** is a collection of open sets which covers a point set.

Definition 7.3 (compactness). A set H is **compact** if for each open covering G of M there is an open covering H with only a finite number of elements such that each element of H is a subset of an element of G . H is called a **finite refinement** of G .

7.1 Exercises

1. If G is a collection of segments of x -axis covering a segment, does a finite subcollection of G cover the segment?
2. If G is a collection of segments of x -axis covering an interval, does a finite subcollection of G cover the interval?
3. If G is a collection of intervals of x -axis covering an interval, does a finite subcollection of G cover the interval?
4. Is the interior of a circle compact?
5. Show that each finite set is compact.

7.2 Theorems

Prove the following.

40. An interval is compact. (Use the axiom.)
41. Each compact set is sequentially compact.

- 42. (Heine-Borel theorem). Each sequentially compact set is compact.
- 43. No interval is the sum of a countable number of closed sets no one of which contains an interval.

In certain abstract spaces, each compact set is sequentially compact but not conversely. In such spaces, Theorem 41 holds but 42 does not. In general, compactness is a stronger condition than sequential compactness and a person who uses it in the hypothesis of a theorem is not proving as strong a result as one who uses mere sequential compactness instead. However, this is not true in the plane. Some important generalizations of Theorem 43 are treated in Chapter 17 on the Baire property.

Chapter 8

Converging Sequences

Definition 8.1. A sequence of points $\{p_1, p_2, \dots\}$ (in which the p_i 's do not have to be distinct) is said to **converge** to p (written $\lim p_n = p$) provided that for each neighborhood D of p there is an integer n such that $p_n + p_{n+1} + \dots$ lies in D .

Definition 8.2. A sequence of numbers $\{A_1, A_2, \dots\}$ is said to **converge** to A provided that for each positive number ε there is an integer n such that no term of $\{A_n, A_{n+1}, \dots\}$ differs from A by more than ε .

Definition 8.3 (Cauchy Sequence). A sequence of points $\{p_1, p_2, \dots\}$ is called a **Cauchy sequence** if for each positive number ε there is an integer n such that no two of the points $\{p_n, p_{n+1}, \dots\}$ are at a distance from each other of more than ε . A Cauchy sequence of numbers is defined in a similar fashion.

Definition 8.4 (Limits). The **upper limit** for a set of numbers is the greatest possible limit of a sequence of numbers in the set. The **lower limit** of a set of numbers is defined similarly.

Definition 8.5 (Distance between sets). The distance between two sets is the greatest lower bound of the distances between the points of one set and the points of the other.

8.1 Exercises

1. Name a set of numbers with no upper limit.
2. Give two mutually exclusive closed point sets whose distance apart is 0.

8.2 Theorems

Prove the following.

44. Every Cauchy sequence of points on the x -axis converges.
45. Each Cauchy sequence of points in the plane converges.
46. Each bounded monotone increasing sequence of numbers converges.
47. Each bounded infinite sequence of points has a converging subsequence.
48. Each bounded infinite collection of numbers has an upper limit and a lower limit.
49. If X, Y are two bounded closed sets, there are points x, y of X, Y respectively such that $\rho(X, Y) = \rho(x, y)$.
50. If X, Y are mutually exclusive closed and bounded sets, they lie at a positive distance from each other.

Chapter 9

Continuous Functions

Definition 9.1. y is a **function** of x on the domain D , written

$$y = f(x) \ (x \in D),$$

provided that for each element of D there corresponds a number $f(x)$. We do not suppose that D is connected or that $f(x)$ is continuous. However, we do suppose that $f(x)$ is single-valued.

Definition 9.2. A point set G is a **graph** of some function $y = f(x) \ (x \in D)$ if it is the collection of all points p such that for some x of D , the coordinates of p are $(x, f(x))$. No vertical line intersects G in two or more points.

Definition 9.3 (Domain of definition). The **domain** of a function is the set of all values x for which the function is defined. It may be considered as the projection of the graph of the function on the x -axis. If the domain of definition of $f(x)$ is D , we may write $f(x) \ (x \in D)$.

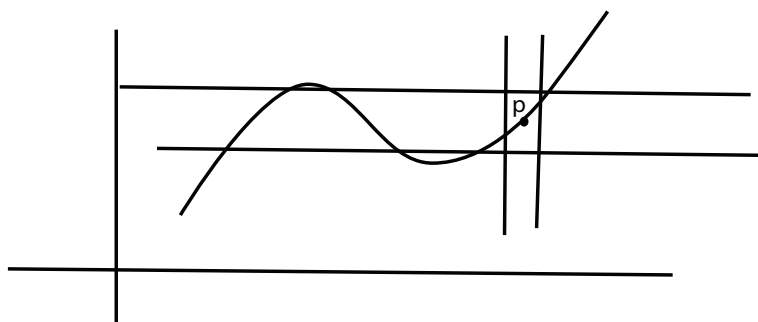
Definition 9.4 (Range of dependent variable). The **range** of $f(x) \ (x \in D)$ is the set of all values y such that for some value x of D , $f(x) = y$. It may be considered as the projection of the graph of the function on the y -axis.

Definition 9.5. A function is said to be **bounded** if the range of the dependent variable is bounded.

Definition 9.6 (Continuity of a graph). The graph G of a function is **continuous** at the point p of G if for each pair of horizontal lines, one above and the other below p , there is a pair of vertical lines, one to the left and one to the right of p such that every point of G between the vertical lines is between the horizontal lines. See Figure 9.1.

We give two definitions of continuity of a function at a point.

Definition 9.7 (Neighborhood definition of continuity at a point). The function $f(x) \ (x \in D)$ is continuous at the value x_0 of X if for each neighborhood N of $f(x_0)$, there is a neighborhood U of x_0 such that $f(y) \in N$ if $y \in U \cap D$.

Figure 9.1: The Continuity of the Graph G at p .

Definition 9.8 (ϵ , δ definition of continuity at a point). The function $f(x)$ ($x \in D$) is continuous at the value x_0 of D if for each positive number ϵ there is a positive number δ (δ is a function of x_0 and f) such that if y belongs to D and differs from x_0 by less than δ , then $|f(y) - f(x)| < \epsilon$.

Definition 9.9 (Continuity of a function). A function is continuous if it is continuous at each value of its domain of definition.

Definition 9.10 (Discontinuity). A function is *discontinuous* if it fails to be continuous at some value of its domain. (Some calculus books call a function discontinuous if its domain of definition is less than all the reals, but we do not follow this convention.)

9.1 Theorems

Prove the following.

51. A continuous function on a sequentially compact domain is bounded.
52. A continuous function on a sequentially compact domain assumes its maximum.
53. The graph of a continuous function on a closed domain is closed.
54. If $f(x)$ and $g(x)$ are continuous functions on the same domain of definition D , then $f(x) + g(x)$ is continuous on D .
55. If $f(x)$ ($0 \leq x \leq 1$) is discontinuous at each irrational value, it is discontinuous at some rational value.
56. Show that if $f(x)$ ($a \leq x \leq b$) and $g(x)$ ($a \leq x \leq b$) are continuous, $f(a) < g(a)$, and $g(b) < f(b)$, then for some value c between a and b , $f(c) = g(c)$.
57. The range of a continuous function is connected if its domain of definition is connected.

58. If the graph of a bounded function is closed, the function is continuous.
59. If $f(x)$ and $g(x)$ are continuous functions on the same domain of definition D , then $f(x) \cdot g(x)$ is continuous on D .
60. If $f(x)$ and $g(x)$ are continuous functions on the same domain of definition D and 0 does not belong to the range of $g(x)$, then $f(x)/g(x)$ is continuous on D .
61. Any subset of a continuous graph is continuous.
62. If $f(x)$ ($a \leq x \leq b$) is continuous and $f(a) < N < f(b)$, there is a value c such that $f(c) = N$. In fact, there is a least such value c .

9.2 Exercises

1. Describe a function that is not continuous anywhere.
2. Describe a function that is continuous at only one point.
3. Describe a function $f(x)$ ($0 \leq x \leq 1$) that is continuous at the irrational values and discontinuous at the rational values.
4. Suppose $F(x,y)$ is a function defined over a plane domain. Give two definitions of what would be meant by $F(x,y)$ being continuous at a point of this domain, one corresponding to each of the definitions above.
5. If $f(x)$ ($0 < x < 1$) is continuous, does it assume its maximum?
6. Give an example of a bounded function whose graph is not bounded.

Chapter 10

Uniform Continuity

Definition 10.1 (Uniform continuity). The function $f(x)$ ($x \in D$) is **uniformly continuous** if for each positive number ε there is a positive number δ such that if x and y are numbers in D differing by less than δ , then $|f(x) - f(y)| < \varepsilon$.

We note that in the above definition the δ is not a function of any particular value x_0 of the domain of definition.

10.1 Exercises

1. Give an example of a function that is continuous but not uniformly continuous.
2. Give an example of a bounded function that is continuous but not uniformly continuous.

10.2 Theorems

Prove the following.

63. If G is an open covering of a sequentially compact set X , there is a positive number $\varepsilon(G)$ such that for each point x of X , $N(x, \varepsilon)$ lies in an element of G . (This number ε is called the Lebesgue number of the open covering G .)
64. A continuous function on a sequentially compact domain of definition is uniformly continuous.

Chapter 11

Sequences of Functions

Definition 11.1 (Convergence). A sequence of functions $\{f_1(x), f_2(x), \dots\}$ all defined on the same domain D is said to converge to a function $f(x)$ on D provided that for each value x of D , $\{f_1(x), f_2(x), \dots\}$ converges to $f(x)$.

Definition 11.2 (Equicontinuity). The sequence of functions

$$\{f_1(x), f_2(x), \dots\}$$

on D is said to be equicontinuous if for each positive number ε there is a positive number δ such that if x, y are values of D differing by less than δ , then for each integer i , $|f_i(x) - f_i(y)| < \varepsilon$.

11.1 Exercises

1. What is meant by saying that a collection of functions is equicontinuous?
2. Give an example of a sequence of uniformly continuous functions that converges to a function that is not continuous.

11.2 Theorems

Prove the following.

65. If a sequence of equicontinuous functions converges to a function $f(x)$, then $f(x)$ is uniformly continuous.
66. If $\{f_1(x), f_2(x), \dots\}$ is a bounded (there is a number N such that $-N < f_i(x) < N$) equicontinuous sequence of functions on the same domain of definition, then some subsequence of $\{f_1(x), f_2(x), \dots\}$ converges to a continuous function on this domain.

Chapter 12

Slopes

Calculus is sometimes called a study of certain aspects of the topology of the real numbers. Functions are certain types of transformations, while derivatives and integrals are types of limits.

If $f(x)$ gets close to b as x gets near but not equal to a we say that $f(x)$ approaches b as x approaches a . More precisely, we give the following definition.

Definition 12.1 (Limit).

$$\lim_{x \rightarrow a} f(x) = b$$

means that a is a limit point of the domain D of f , and for each neighborhood N of b there is a deleted neighborhood U of a such that if $y \in D \cap U$, then $f(y) \in N$.

Definition 12.2 (Slope). Suppose G is the graph of a function, $M(x_p, x_q)$ denotes the slope of the secant line between $p = (x_p, f(x_p))$ and $q = (x_q, f(x_q))$, and p is a point of G which is a limit point of G . Then if

$$\lim_{x_q \rightarrow x_p} M(x_p, x_q)$$

exists, this limit is called the slope of G at p .

Definition 12.3 (Derivative). Suppose a is a point and a limit point of the domain of definition D of $f(x)$. If

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

then we call $f'(a)$ the **derivative** of f at a .

Disregarding the domain of definition of $f(x)$, as is often done in casual treatments of calculus, we have that

$$f'(a) = \lim_{\varepsilon \rightarrow 0} \frac{f(a + \varepsilon) - f(a)}{\varepsilon}$$

means that for each neighborhood N of $f'(a)$ there is a deleted neighborhood U of 0 such that $\frac{f(a+h)-f(a)}{h}$ is an element of N if h is in U .

Definition 12.4. A graph G is **rising** at the point p of G if there exist two vertical lines, one to the left and the other to the right of p such that every point of G between these vertical lines is higher or lower than p according as it is to the right or left of p .

12.1 Exercises

1. Give an ε, δ definition of what is meant by $\lim_{x \rightarrow a} f(x) = b$.
2. Give an ε, δ definition of what is meant by a function rising.
3. In the definition of limit, is it concluded that $f(a) = b$?
4. Make a definition of what would be meant by saying that $\lim_{x \rightarrow \infty} f(x) = b$.
5. Give an example of a function $f(x)$ ($0 \leq x \leq 1$) that has a derivative everywhere but whose derivative is not continuous at $x = 0$.
6. Give an ε, δ definition of a derivative.

12.2 Theorems

Prove the following.

67. If a function $f(x)$ ($0 \leq x \leq 1$) is rising at every point, it is continuous for at least one value.
68. If $f(x)$ ($a \leq x \leq b$) is rising at every point and $a \neq b$, then $f(a) < f(b)$.
69. If a graph G has a slope at a point, it is continuous there.
70. If the graph of $f(x)$ is rising at $(c, f(c))$, then $f'(c) > 0$ if $f'(c)$ exists.
71. If $f(x)$ ($a \leq x \leq b$) has a maximum at a value c between a and b then $f'(c) = 0$ if it exists.
72. If $f'(x)$ ($a \leq x \leq b$) exists and $a \neq b$, then there is a value c between a and b such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.
73. If $f'(x)$ ($a \leq x \leq b$) exists and $f'(a) < N < f'(b)$, then there is a value c between a and b such that $f'(c) = N$.

Chapter 13

Transformations

Definition 13.1. A **transformation** f of X into Y is a function, rule, or law which assigns to each element x of X a unique element $f(x)$ of Y .

Definition 13.2. If for each point y of Y there is at least one element x of X such that $f(x) = y$, the transformation is said to take X **onto** Y .

Definition 13.3. If for each y of Y there is exactly one element x of X such that $f(x) = y$, the transformation is called **one-to-one** or (1-1).

Equations are examples of transformations of the real numbers into the real numbers. $f(x) = x^3$ is both (1-1) and onto, $f(x) = x^3 - x$ is onto but not (1-1), while $f(x) = x^2$ and $f(x) = \sin(x)$ are neither (1-1) nor onto.

Rotations, translations, projections, congruences, and similarities are examples of transformations encountered in geometry.

Also, real valued functions may be regarded as transformations of points on the x -axis into points of the plane on the same vertical line with them. Hence $f(x) = x^2$ ($1 < x < 2$) may be regarded as a transformation of the part of the x -axis between $(1,0)$ and $(2,0)$ onto the part of the graph of $y = x^2$ between $(1,1)$ and $(2,4)$.

Furthermore, a real valued function may be regarded as a transformation on the x -axis. Then $f(x) = x^2$ would take the point $(a,0)$ into the point $(0,a^2)$. See Figure 13.1.

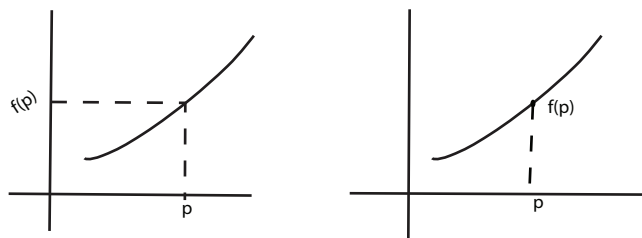


Figure 13.1: Transformations on the x -axis.

Definition 13.4. If f is a transformation of X into Y and A is a subset of X , then the **image** of A , denoted $f(A)$, is the set $\{f(x) \mid x \in A\}$. See Figure 13.2 below.

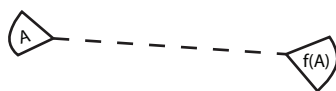


Figure 13.2: A and its image $f(A)$.

Definition 13.5. If $B \subset f(X)$, the **inverse** of B , denoted $f^{-1}(B)$, is the set $\{x \mid f(x) \in B\}$.

As an example of $f^{-1}(B)$, consider the example shown in Figure 13.3, in which the function f is a projection onto a line.

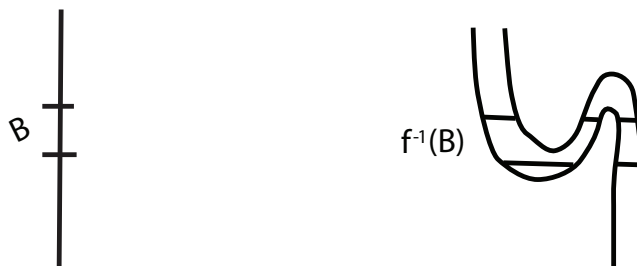


Figure 13.3: f is a Projection onto a line.

Definition 13.6 (Neighborhood definition of continuity). A transformation T of X into Y is **continuous at the point** x of X if for each neighborhood N of $f(x)$ there is a neighborhood $U(x, N)$ of x such that $T(U \cdot X) \subset N$. T is **continuous** if it is continuous at each point of X .

Definition 13.7. A continuous transformation is called a **mapping**.

Definition 13.8. The image of an interval under a mapping is called a **continuous curve**. (Also called a Peano curve.)

13.1 Exercises

1. Give an example of an into transformation that is not an onto transformation.
2. Describe a 1-1 transformation of the non-negative real numbers onto the reals.
3. Is there a transformation of the plane onto a point? Of a point onto the plane?

4. Give an ε, δ definition of continuity of a transformation.
5. Give a definition of uniform continuity of a transformation.
6. Show that a continuous transformation does not preserve any of the following properties individually:
 - (a) boundedness,
 - (b) being closed,
 - (c) being infinite,
 - (d) being unbounded,
 - (e) having area of 1 square foot,
 - (f) having length of 1 inch.
7. Suppose T is a continuous transformation of the plane into the plane. Which of the following properties implies that it is continuous at the origin?
 - (a) T is continuous on each line through the origin.
 - (b) T is continuous on each vertical and horizontal line.
 - (c) T is continuous on each straight line.
 - (d) T is equicontinuous on the lines through the origin.
 - (e) T is equicontinuous on all vertical lines and continuous on horizontal lines.

13.2 Theorems

Prove the following.

74. Connectedness is preserved by a continuous transformation.
75. Sequential compactness is preserved by a continuous transformation.
76. If T is a continuous transformation of X , then for each relatively open subset N of $T(X)$, $T^{-1}(N)$ is a relatively open subset of X .
77. A transformation is continuous if the inverse of open sets are open.
78. A square plus its interior is a continuous curve.
79. If x is a limit point of M and T is a continuous transformation of M , then $T(x)$ is either a point or a limit point of $T(M)$.
80. If $\{x_1, x_2, \dots\}$ converges to x_0 , then $\{T(x_1), T(x_2), \dots\}$ converges to $T(x_0)$ if T is a continuous transformation defined on a set containing the x 's.

Chapter 14

Function of a Function

If g is a map of X into Y and f is a map of Y into Z , then fg is a map of X into Z . To find $fg(x)$, first consider x , then its image $g(x)$ under g , and finally the image $fg(x)$ of $g(x)$ under f . See Figure 14.1.

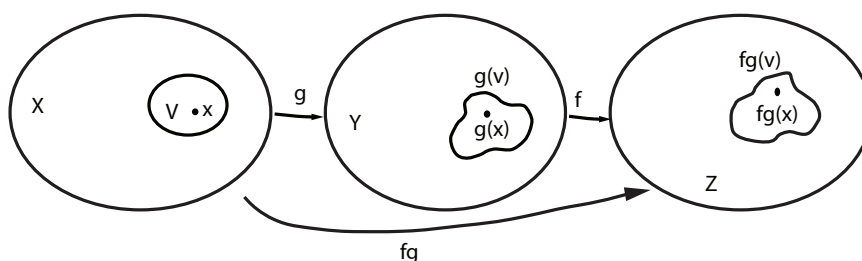


Figure 14.1: Function of a Function.

To show that fg is continuous at x , we need to show that for each neighborhood U of $fg(x)$ there is a neighborhood V of x such that $fg(V \cdot X) \subset U$. This is done by considering a neighborhood W of $g(x)$ such that $f(W \cdot Y) \subset U$ and then picking V so that $g(V \cdot X) \subset W$.

If f and g are real valued functions of real numbers, the graph of fg may be constructed from the graphs of f and g .

Here is how to plot $(a, fg(a))$. Using a , find the value of $g(a)$ on the graph of g . Using this value $g(a)$ thus obtained, find the value $f(g(a))$ on the graph of f . Then plot the point $(a, fg(a))$ on the graph fg .

In Figure 14.2, we let $g(x) = x^2 - 4$, $f(x) = \sin(x) - x$, and $fg(x) = \sin(x^2 - 4) - (x^2 - 4)$.

Of course in plotting, it is much better to get an idea of the general position of many points at once rather than plot them one at a time.

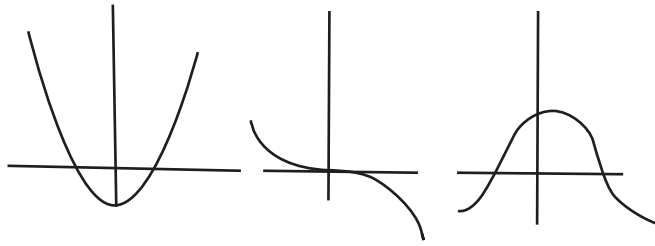


Figure 14.2: Several Graphs.

14.1 Theorem

Prove the following:

81. If g is a continuous transformation on X and f is a continuous transformation of $g(X)$, then fg is continuous on X .

Chapter 15

Homeomorphisms

Definition 15.1. If f is a (1-1) transformation of X onto Y , f^{-1} is a (1-1) transformation of Y onto X . If both f and f^{-1} are continuous, f is called a **homeomorphism**.

Not every (1-1) continuous transformation is a homeomorphism. For example, let $f(x)$ be the function defined for $x \geq 0$ by

$$f(x) = \left(\cos \left(\frac{2\pi x}{x+1} \right), \sin \left(\frac{2\pi x}{x+1} \right) \right).$$

$f(x)$ takes a ray continuously and (1-1) onto a circle in the plane, but f^{-1} is not continuous at $(1, 0)$.

Definition 15.2. Two sets are **homeomorphic** or **topologically equivalent** if there is a homeomorphism of one onto the other.

Definition 15.3. Any set topologically equivalent to an interval is called an **arc**.

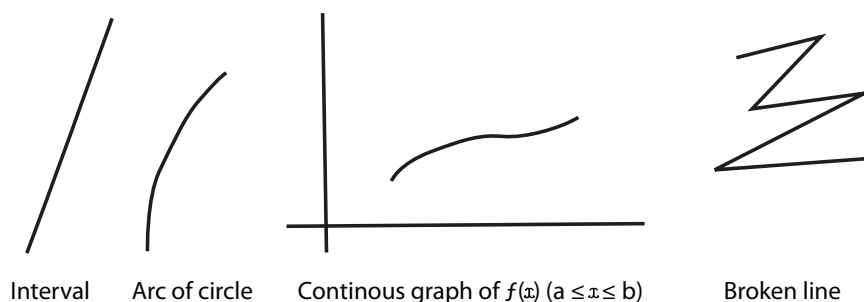


Figure 15.1: Some Arcs.

Definition 15.4. Any set topologically equivalent to a circle is called a **simple closed curve** or a **1-sphere**.

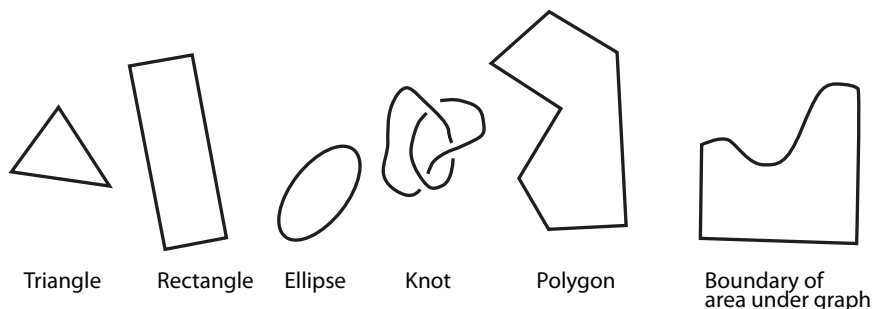


Figure 15.2: Some Simple Closed Curves.

It may be shown that in the plane the “interior” of any simple closed curve is topologically equivalent to the interior of a circle. This is part of the Jordan curve theorem.

When one speaks of the area of a circle, he probably refers to the area of its interior. There are simple closed curves that wiggle around so much that in a certain sense, they have no area. However, it is of importance in integral calculus that if G is the continuous graph of $y = f(x)$ ($a \leq x \leq b$) and $f(x) > 0$ and J is the simple closed curve which is the sum of G and the intervals from $(a, 0)$ to $(a, f(a))$, $(a, 0)$ to $(b, 0)$, and $(b, 0)$ to $(b, f(b))$, then the area of the interior of J does exist. It is called $\int_a^b f(x) dx$.

Definition 15.5. A set topologically equivalent to the surface of a sphere is called a *simple surface* or a *2-sphere*.

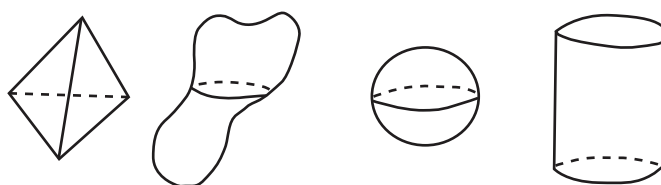


Figure 15.3: Some Simple Surfaces.

There are wild simple surfaces in Euclidean 3-space whose interiors are not topologically equivalent to the interior of a sphere.

Although two surfaces are topologically equivalent to each other it may not be possible to push, pull, and stretch one without breaking and tearing to make it fit on the other. See Figure 15.4 for a picture of two such surfaces.

Definition 15.6. A set is *nondegenerate* if it contains more than one element.

Definition 15.7. A *continuum* is a closed and connected point set.

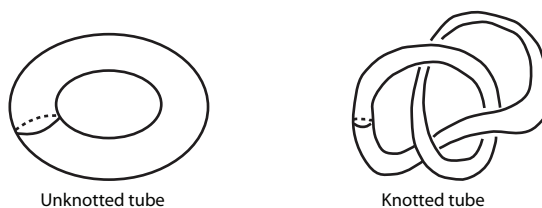


Figure 15.4: Topologically Equivalent, but

15.1 Exercises

1. Describe a homeomorphism between a disk and a triangle plus its interior.
2. Show that a simple closed curve is bounded.
3. Show that an arc is not homeomorphic with a simple closed curve.
4. Let M be the sum of the graph of

$$y = \sin\left(\frac{1}{x}\right) \quad \left(-\frac{1}{\pi} \leq x \leq \frac{1}{\pi}, x \neq 0\right)$$

and the origin. Is M connected? Is M a continuum? Is there a (1-1) continuous transformation of M onto an interval?

15.2 Theorems

Prove the following.

82. A segment is homeomorphic with a line.
83. The plane is homeomorphic with the interior of a circle.
84. The surface of a sphere minus a point is homeomorphic with the plane.
85. An open subset of the plane is connected if and only if each pair of its points can be joined by an arc in it.
86. The homeomorphic image of a nondegenerate sequentially compact continuum is a nondegenerate sequentially compact continuum.
87. If p and q are two points of a simple closed curve J , then $J - (\{p\} + \{q\})$ is not connected.
88. If p is a non end point of an arc A , then $A - \{p\}$ is not connected.
89. There is an arc with no length.
90. The graph of a continuous function $f(x)$ ($a \leq x \leq b, a < b$) is an arc.

Chapter 16

Cantor Set

Definition 16.1. Let $M_1 = [0,1]$ be a straight line interval, $M_2 = M_1 - (1/3, 2/3)$ be the part of M_1 remaining after the open middle third of it is removed, $M_3 = M_2 - (1/9, 2/9) - (7/9, 8/9)$ be the part remaining after the open middle thirds of the intervals in M_2 are removed, $\dots$ and so on. The intersection of $M_1, M_2, \dots$ is called the **Cantor set** or the **Cantor discontinuum** or the **Cantor ternary set**.

Definition 16.2. A set is **perfect** if it is closed and every point of it is a limit point of it.

The Cantor set is an example of a perfect set that contains no nondegenerate continuum.

16.1 Exercises

1. Find the sum of the lengths removed in forming the Cantor set.
2. If at the first step the middle open third of $[0,1]$ was removed, at the next step the open middle ninths of the remaining intervals were removed, at the next step the open middle twenty sevenths were removed, $\dots$, show that the sum of the lengths removed is different from 1. Show that the resulting set is homeomorphic with a Cantor set.

16.2 Theorems

Prove the following.

91. An interval is the image of the Cantor set under some continuous transformation. (It can be shown that any sequentially compact set is such an image.)
92. Any perfect set has uncountably many points.

Chapter 17

Baire property

Definition 17.1. A subset M of A is called **dense** in A if each point of A belongs to $\overline{M}$.

Definition 17.2. A set X is **separable** if some countable subset of X is dense in X .

Definition 17.3. A subset M of A is called **nowhere dense** in A if M does not contain any relatively open subset of A .

Definition 17.4. The intersection of a countable number of open sets is called an **inner limiting set** or a G_δ set.

Definition 17.5. The sum of a countable number of closed sets is called an F_σ set.

17.1 Theorems

Prove the following. Theorems 93 and 94 are extensions of Theorem 43.

93. The square plus its interior is not the sum of a countable number of nowhere dense sets.
94. (Baire property theorem.) No sequentially compact set X is the sum of a countable number of closed sets no one of which contains a relative open subset of X .
95. A closed set is an inner limiting set.
96. The complement of a G_δ set is an F_σ set and conversely.
97. No interval is the sum of a countable number more than one of mutually exclusive closed sets.
98. The plane is separable.

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- [1] Parker, John, *R. L. Moore : Mathematician and Teacher*, Washington, DC: Mathematical Association of America, 2005.